

Optical and Acoustic Characterization of Phase-Shift Droplets with Varying Shell Compositions

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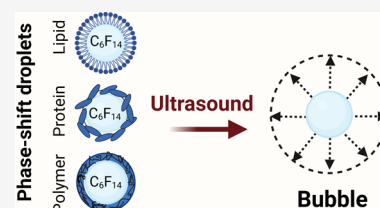
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ABSTRACT: Shell-stabilized, phase-shift droplets of perfluorocarbon have shown promising potential in diagnostic and therapeutic ultrasound applications. While the role of shell composition has been well-studied for conventional gas-filled microbubbles, its influence on bubbles generated via acoustic droplet vaporization (ADV) remains understudied. This study investigates the effect of shell composition—lipid, protein, and polymer—on the stability, ADV dynamics, growth behavior, release kinetics, and acoustic response of perfluorohexane phase-shift droplets. Payload-carrying, micron-sized droplets were produced using a microfluidic platform. Both optical, including ultrahigh-speed and timelapse confocal microscopy, and acoustic characterizations of the generated droplets were investigated in fibrin-based, tissue-mimicking hydrogels. Fluorescent markers were used to selectively label fibrin, payload, and droplet shells. Following insonation at 2.5 and 9.6 MHz, lipid-coated droplets had a maximum expansion ratio up to 20% lower than protein- and polymer-coated droplets. Payload release rates were an order of magnitude faster than the bubble growth rate post ADV, with minimal dependence on shell composition. Additionally, lipid shell fluorescence retained up to 40% of its initial intensity 120 s after ADV, while protein shell fluorescence was completely diminished. Acoustic characterization indicated that the ADV and inertial cavitation thresholds were similar across all shell types, suggesting that shell composition has a negligible effect on acoustic characteristics at high driving pressures.



INTRODUCTION

Liquid perfluorocarbon droplets, also known as phase-shift droplets, can be selectively vaporized into echogenic gas bubbles in situ by externally applied ultrasound,¹ enabling a range of diagnostic and therapeutic applications.^{2,3} The stability of the phase-shift droplets depends on several factors, including the properties of the encapsulating shell, the type of liquid core, and the characteristics of the acoustic waveform. The encapsulating shell plays a crucial role in stabilizing conventional gas-filled microbubbles, preventing coalescence, extending their lifespan by limiting gas dissolution, and minimizing active clearance by the immune system.^{4,5} Additionally, shell composition is a key determinant in modulating the nonlinear oscillation of conventional microbubbles.^{6,7} Common materials for shell encapsulation include proteins, phospholipids, and synthetic polymers. The thickness of these shells varies significantly from 1 to 5 nm in lipid shells, to 15–20 nm in protein shells, and up to 100–200 nm in polymer shells.⁸ Biochemical analyses suggest that protein and lipid shells typically form monomolecular layers at the air–water interface, whereas polymer shells are generally thicker due to covalent cross-linking and polymeric entanglement.^{8,9} While it is possible to modulate the thickness-to-radius (T/R) ratio by adjusting the polymer-to-perfluorocarbon ratio, the smallest ratio achieved was approximately 0.3 as measured by confocal microscopy.¹⁰ However, modifying the shell thickness

would require corresponding changes in particle size, which significantly impacts their radial dynamics and acoustic emissions.

Soft-shell materials, such as lipids, generally allow greater bubble expansion and contraction, whereas hard-shell materials, such as proteins and polymers, tend to restrict oscillatory behavior due to their rigidity. Existing literature has documented the impact of shell composition on the dynamics and acoustic responses of conventional microbubbles.^{11–14} For example, protein and polymer shells significantly dampen oscillations and exhibit cracking under high acoustic pressures.¹⁵ Lipid shells, in contrast, consist of a self-assembled monomolecular layer held together by hydrophobic forces and other physical associations (e.g., intermolecular dispersion and electrostatic forces), rather than covalent bonding or chain entanglement as seen in protein and polymer shells.^{8,16,17} This structure allows the lipid shell to rapidly reseal following dilatation or fracture, minimizing surface tension and stabilizing the gas core.^{4,17,18} Phospholipid-based conventional

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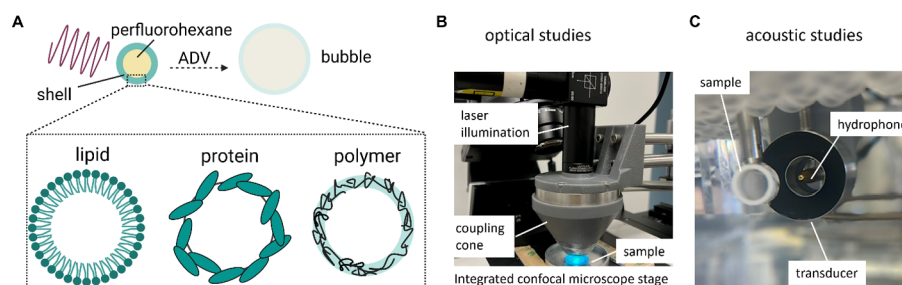


Figure 1. Experimental setups for optical and acoustic characterization of phase-shift droplets stabilized with lipid, protein, and polymer shells. (A) Illustration of acoustic droplet vaporization (ADV), where phase-shift droplets with varying shells are converted into bubbles upon ultrasound exposure. Panel A was created with BioRender. (B) Integration of ultrahigh-speed microscopy and confocal microscopy with focused ultrasound to investigate ADV dynamics during and up to 120 s after ultrasound activation. (C) Acoustic emissions from the droplets were captured using a hydrophone in passive mode to analyze their acoustic response.

microbubbles, although generally less stable than protein and polymer shells, are frequently studied due to their high acoustic responsiveness and echogenicity.⁴ While the effects of shell composition on gas-filled microbubble dynamics are well-documented, relatively little research has focused on how these factors influence bubbles formed through acoustic droplet vaporization (ADV) (Figure 1A). Understanding the role of shell material in ADV is important for gaining insights into processes such as bioeffects, drug delivery kinetics of shell-bound therapeutic agents, scattering efficiency, and the long-term stability of the generated bubbles. This knowledge can inform the design of phase-shift droplets tailored for specific diagnostic and therapeutic applications.

ADV-induced bubbles undergo significant radial expansions due to the phase-change process, which depends on factors such as excitation frequency, droplet size, and surrounding mechanical properties.^{19–21} For example, a 12- μm phase-shift droplet with a perfluoropentane core can achieve maximum expansion ratios of approximately 5 and 9 at 2.5 and 1 MHz, respectively, at the peak rarefactional pressure of 4 MPa.²⁰ In the context of conventional gas-filled microbubbles, the extent to which the microbubbles expand and contract during rarefaction and compression is significantly smaller.²² For instance, a 2- μm microbubble expands in diameter by approximately 2.5 times during rarefaction at 2.4 MHz and 400 kPa.⁵

Research on monodisperse microbubbles has highlighted the benefits of a controlled size distribution, demonstrating significant improvements in the signal-to-noise ratio of contrast-enhanced ultrasound imaging compared to a polydisperse population. For example, acoustically sorted microbubbles have shown a two-orders-of-magnitude increase in sensitivity (measured as the average scattering cross-section per bubble) compared to native polydisperse Bracco bubbles (Bracco BR-14, Bracco Research, Geneva, Switzerland).²³ The enhanced sensitivity of a monodisperse population can be of great interest not just for imaging but for therapeutic applications like drug delivery where typically small amounts of agents are retained at the target site. Additionally, sonoporation efficiency has been shown to be highly size-dependent, with polydisperse microbubbles exhibiting an order of magnitude lower sonoporation efficiency than their monodisperse counterparts.²⁴

To achieve narrow-size distributions of phase-shift droplets, similar to those in conventional microbubbles, techniques like filtration, centrifugation, and microfluidics have been explored.³ Microfluidic techniques, in particular, have enabled

the controlled production of monodisperse, micron-sized (1–15 μm)^{25–27} and nanosized (200–500 nm)²⁸ droplets, resulting in uniform ADV threshold and improve vaporization efficiency. Various shell materials have been investigated to enhance droplet stability and in vivo performance, including lipids,²⁹ poly(DL-lactide-co-glycolide),³⁰ Pluronic F68,³¹ bovine serum albumin,¹ and fluorinated surfactants.²⁸ Phospholipid- and albumin-stabilized droplets are commonly studied for diagnostic applications, whereas polymer-stabilized droplets show potential for therapeutic applications, such as drug delivery, due to their mechanical robustness and high payload capacity. Recently, hybrid-shelled microdroplets have been introduced, where the positive surface potential provided by a long acyl chain gel-crystalline surfactant ensures high colloidal stability. The additional polymer layer enhances robustness and provides elastomeric properties, creating a versatile system with potential advantages for biomedical applications.^{32,33} Furthermore, studies have examined factors influencing droplet size and stability, such as flow rate ratios in microfluidic devices.^{27,28} For example, polymeric and fluorinated surfactant shells have been found to enhance storage stability for perfluorocarbon nanodroplets compared to lipid shells, which exhibit higher polydispersity due to vesicle aggregation.²⁸ Shell composition has been shown to affect stability of droplets in vivo as well, with longer acyl chains in phospholipid shells enhancing droplet stability up to 20-fold.³⁴ However, experimental research on how shell material specifically impacts the radial dynamics and acoustic emissions of monodisperse phase-shift droplets remains unexplored.

In this study, we characterized phase-shift droplets, generated via microfluidic techniques and stabilized by three different shell materials—polymer, phospholipids, and protein—using both optical and acoustic analyses. Lipids, proteins, and polymers were selected as representative shell materials for phase-shift droplets because they are widely used in biomedical applications of conventional gas-filled microbubbles. Examples include FDA-approved formulations such as Definity (Lantheus, lipid-based), Sonovue (Bracco, lipid-based), Optison (GE Healthcare, protein-based), and Albutex (protein-based).⁵

We used ultrahigh-speed microscopy at 10 million frames per second to provide quantitative information on the dynamics of ADV-generated bubbles within fibrin-based, tissue-mimicking hydrogels. Furthermore, time-lapse confocal microscopy enabled real-time assessment of bubble growth, payload release kinetics, and shell retention. Finally, we compared the acoustic emissions, both linear and nonlinear

responses, to understand how different shell materials affect ADV-bubble behavior.

MATERIALS AND METHODS

Preparation of Phase-Shift Droplets Using a Microfluidic Platform. Phase-shift droplets with a perfluorohexane (PFH; C_6F_{14} ; Strem Chemicals, Newburyport, MA, USA) liquid core were fabricated using a microfluidic platform. Higher bulk boiling point perfluorocarbons, such as PFH, are used to minimize spontaneous vaporization (i.e., vaporization in the absence of ultrasound), making them well-suited for applications such as drug delivery and tissue regeneration. These droplets had a double emulsion structure similar to our previous studies.^{25,35} To enable visualization and facilitate the study of ADV-induced payload release, Cascade blue dextran (Thermo Fisher Scientific, Waltham) was added to the PFH solution at a concentration of 1 mg/mL in phosphate-buffered saline (PBS, pH 7.4).

Lipid-Stabilized Droplets. The primary phospholipid component used for the encapsulation was 1,2-dipalmitoyl-*sn*-glycero-3-phosphocholine (DPPC). Two formulations were prepared. In one formulation, DPPC was combined with 1,2-dipalmitoyl-*sn*-glycero-3-phosphate (DPPA) at a 9:1 molar ratio. DPPA provided negative charges, enhancing the zeta potential and stability of the droplets. The second formulation included DPPC, DPPA, and 1,2-dipalmitoyl-*sn*-glycero-3-phosphoethanolamine-N-[methoxy (polyethylene glycol)-5000] (DPPE-PEG-5k), similar to the shell composition of the commercially available Definity contrast agent (Lantheus Medical Imaging, Billerica, MA, USA).³⁶ Long-chain PEG brushes are commonly used to reduce coalescence through steric repulsion and to minimize clearance by macrophages.³⁷ Lipid solutions were prepared using the thin film hydration technique.²⁹ Lipid films were rehydrated with an aqueous buffer containing 80% (v/v) PBS, 10% (v/v) propylene glycol, and 10% (v/v) glycerol, with a final lipid concentration of 10 mg/mL. Dispersion and mixing of the lipids were facilitated by heating the solution to 10 °C above the main phase transition temperature (T_c) of DPPC (T_c : 41 °C). All lipids were obtained from Avanti Polar Lipids (Alabaster, AL, USA) and stored at −20 °C prior to use.

Protein-Stabilized Droplets. For protein encapsulation, bovine serum albumin (BSA) was utilized at concentrations ranging from 4 to 30 mg/mL to evaluate the effect of protein concentration on droplet stability. Dextrose (50 mg/mL) was added to increase viscosity, aiding in droplet stabilization during microfluidic production.³⁸ All chemical reagents used for the preparation of protein-stabilized droplets were purchased from Sigma-Aldrich (St. Louis, MO, USA).

Polymer-Stabilized Droplets. Polymer-stabilized droplets were generated using a 50 mg/mL solution of Pluronic F68 (CAS# 9003-11-6, Sigma-Aldrich), a block copolymer surfactant, following protocols from previous publications.^{20,35,39}

Microfluidic Production. Droplets were produced using a microfluidic system equipped with a flow-focusing junction (14 μm $\times$ 17 μm , Dolomite, Royston, UK). The system was maintained at room temperature, below the bulk boiling point of PFH. Phase-shift droplets with an approximate diameter of 12 μm were produced by setting the flow rates of the dispersed (perfluorocarbon) and continuous (shell) phases to 0.5 and 2.5 $\mu\text{L}/\text{min}$, respectively. This specific flow rate ratio was selected based on our prior studies.^{20,39} The larger droplet size was selected for its relevance to our previous studies and its utility in research focusing on tissue regeneration and drug delivery applications. During production, droplet formation was monitored in real-time using an inverted microscope (Leica Microsystems, IL, USA) coupled with a digital camera (Teledyne Technologies Inc., CA, USA). All droplets were stored at 4 °C and used within 45 days. In this study, we only labeled lipid and BSA shells. The limited use of fluorescent labeling for polymeric shells, compared to BSA and lipid shells, is primarily due to the inherent lack of reactive functional groups in many polymers. This absence necessitates additional chemical modifications to introduce functional

groups that can react with fluorescent dyes, complicating the labeling process.⁴⁰ To prepare fluorescently labeled phospholipid shells, a lipophilic fluorescent marker DiD (Invitrogen, Waltham, MA, USA) was added to the droplet suspension following a previous study.⁴¹ BSA was labeled with Alexa Fluor 647 conjugate (Invitrogen). Encapsulation efficiency of payload in droplets with varying shell materials was measured, as done previously,⁴² by breaking the emulsion using a probe sonicator (Branson 450A, Emerson, St. Louis, MO, USA).

Size Characteristics of Droplets and Shell Properties. The resulting droplets were measured using a Coulter Counter (Beckman Coulter, Brea, CA, USA) with a 30- μm aperture, which allowed for the measurement of droplets ranging from 0.6 to 18 μm in diameter. The Coulter counter measures both particle count and size distribution using an electrical zone sensing method, which is independent of the particles' optical properties.⁴³ Three independent measurements were taken for each formulation. Additionally, surface tension was measured at room temperature using a Du Nouy ring tensiometer (range: 0–90 mN/m, Swedesboro, NJ, USA).⁴⁴ For each test, 30 mL of the sample was used, and the distance between the immersed ring and the liquid surface was maintained at approximately 3 mm to ensure a clean break of the meniscus on the platinum ring. Surface tension measurements were conducted for three different shell solutions (polymer, lipid, and protein). Similarly, interfacial tension between PFH and the shell solutions was determined using the same instrument to further characterize the interaction between the shell material and the droplet core.

Preparation of Fibrin-Based Hydrogels. Fibrin-based hydrogels were prepared according to previously established protocols to create a tissue-mimicking environment for phase-shift droplets.^{31,35} Bovine fibrinogen (Sigma-Aldrich) was dissolved in FluoroBrite Dulbecco's Modified Eagle Medium (DMEM, Life Technologies) at a concentration of 20 mg/mL. To form the hydrogels, the fibrinogen solution was combined with DMEM, droplets, and human recombinant thrombin (2 U/mL, Recothrom, Baxter, Deerfield, IL, USA), resulting in a final fibrin concentration of 10 mg/mL. For optical studies, including high-speed and time-lapse confocal imaging, a low volume fraction of PFH droplets (0.006% (v/v)) was incorporated to minimize interdroplet interactions. A higher volume fraction of droplets (0.05% (v/v)) was used for acoustic characterization studies to enhance the detection and analysis of acoustic emissions. The acoustic properties of fibrin-based hydrogels containing phase-shift droplets were characterized in a prior study, yielding an attenuation coefficient of $0.14 \pm 0.006 \text{ dB cm}^{-1} \text{ MHz}^{-1}$ and a speed of sound of $1487 \pm 2.3 \text{ m/s}$ (frequency range: 1–12 MHz).⁴⁵ Fibrin was fluorescently labeled with 39 $\mu\text{g}/\text{mL}$ of either Alexa Fluor 647- or Alexa Fluor 488-labeled fibrinogen (Molecular Probes, Eugene, OR, USA) for confocal imaging.

Ultrahigh-Speed Brightfield and Time-Lapse Fluorescence Confocal Microscopy. Bubble dynamics during ADV were visualized using ultrahigh-speed imaging at 10 million frames per second (Mfps). The fibrin-based hydrogels containing phase-shift droplets were polymerized within a custom-made container placed on the stage of a confocal microscope (AX, Nikon, Tokyo, Japan) equipped with a 40x objective (PLAN APO, N.A.: 1.15, Nikon) (Figure 1B). The temperature was maintained at 37 °C using a microscope stage temperature control system (TCS-200, AmScope).

A focused transducer (f-number: 0.83, radius of curvature: 50 mm, H-147, Sonic Concepts Inc., Bothell, WA, USA) was aligned confocally with both the ultrahigh-speed camera (HPV-X2, Shimadzu) and the confocal microscope. Ultrasound exposure was delivered by coupling the focused transducer to the hydrogel through a 3D-printed cone (Form 3L, Formlabs, Boston, USA) filled with deionized water as shown in Figure 1B. Two excitation frequencies of 2.5 and 9.6 MHz were used. At each frequency, a single burst of 6 cycles at a peak negative pressure of 5 MPa was applied. Two ultrasound frequencies, 2.5 and 9.6 MHz, were selected to represent low and high frequencies commonly used in clinical applications. Illumination for imaging was provided by a high intensity pulsed laser (400 W, 640 nm, Cavitar, Tampere, Finland) in a transmission

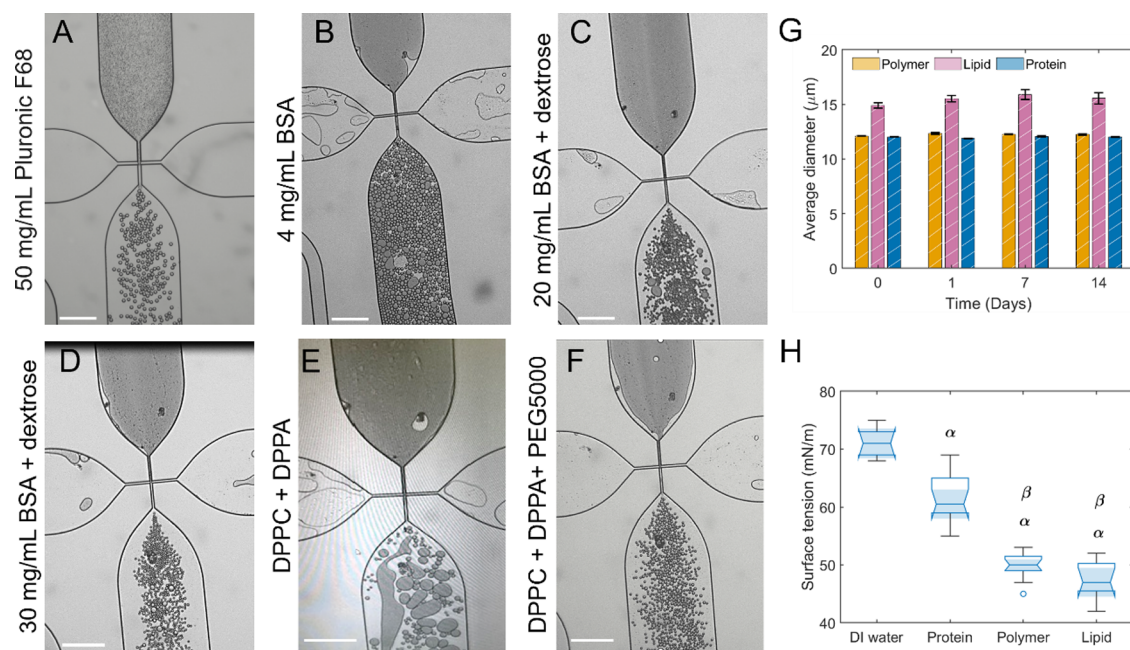


Figure 2. Microfluidic generation and stability assessment of PFH phase-shift droplets with varying shell compositions. (A) Polymer-stabilized droplets produced with 50 mg/mL Pluronic F68. Protein-stabilized droplets with BSA at concentrations of (B) 4 mg/mL, (C) 20 mg/mL with 50 mg/mL dextrose, and (D) 30 mg/mL with 50 mg/mL dextrose. Lipid-stabilized droplets using (E) DPPC combined with DPPA and (F) DPPC + DPPA + PEG5000. (G) Average diameter of polymer-, lipid-, and protein-stabilized droplets over a 14-day storage period, showing size stability for each shell composition ($n = 3$ for each shell composition). (H) Surface tension measurements were compared for four solutions ($N = 8$). Scale bar: 200 μm .

configuration. A complete detail of synchronization can be found in our previous publication.²⁰ To capture the longitudinal growth of ADV-generated bubbles over time and quantify payload release kinetics, confocal fluorescence imaging was performed at a rate of 1 fps for a duration of 120 s.

Acoustic Characterization of Phase-Shift Droplets. ADV was induced using a calibrated focused transducer (Sonic Concepts) operating at 2.5 MHz. Pulsed waveforms (pulse duration: 3 μs and pulse repetition frequency: 100 Hz) were generated by a function generator (33500B, Agilent Technologies, Santa Clara, CA, USA) and amplified by a gated power amplifier (Ritec, GA-2500A, Warwick, RI, USA) at peak negative pressures up to 6 MPa. The acoustic emissions resulting from ADV were recorded passively using a capsule hydrophone (HGL-1000, frequency range: 1–20 MHz, Onda, Sunnyvale, CA, USA)⁴⁶ (Figure 1C). The recorded signals were acquired directly by a digital oscilloscope (sampling rate: 100 MHz, HDO4034, Teledyne LeCroy, Chestnut Ridge, NY, USA) using MATLAB (MathWorks, Natick, MA, USA) and processed offline as outlined previously.⁴⁶ All experiments were conducted in a water tank filled with degassed (<25% dissolved oxygen saturation) DI water at 37 $^{\circ}\text{C}$.

Statistical Analysis. All statistical analyses were performed in MATLAB using one-way ANOVA followed by Tukey's multiple comparisons test. Data is presented in box plots indicating the median, interquartile range, and outliers for each condition. The number of independent replicates (n) is specified in each figure caption. Statistical significance was determined at a confidence level of 95% ($p < 0.05$).

RESULTS AND DISCUSSION

Stability of Monodisperse Phase-Shift Droplets with Varying Shell Compositions. A variety of techniques can produce phase-shift droplets such as agitation, sonication, extrusion, condensation and microfluidic techniques.³ While size isolation techniques such as filtration and centrifugation can separate specific droplet diameters, these processes

typically result in reduced production yields and make it challenging to isolate monodisperse populations. Flow-focusing microfluidic devices, however, enable highly reproducible production of monodisperse phase-shift droplets with tunable sizes, tighter size distributions, and payloads that can be adjusted in real-time by modulating flow rates. Our group primarily has studied monodisperse polymer-coated (e.g., Pluronic F68) droplets due to their excellent stability and reproducible fabrication for tissue regeneration applications. In this work, we extended our study to protein- and lipid-stabilized droplets.

In this study, phase-shift droplets were generated using a microfluidic platform with a flow rate of 2.5 $\mu\text{L}/\text{min}$ for the continuous phase containing the shell material and 0.5 $\mu\text{L}/\text{min}$ for the dispersed phase containing PFH liquid. This flow rate combination follows established methods from our previous publications, which predominantly used Pluronic F68 as the shell material. Polymer-stabilized droplets exhibited a stable flow dynamic during microfluidic production, resulting in a consistent and monodisperse population (Figure 2A). No significant aggregation or coalescence was observed, confirming the robustness of Pluronic F68 as a stabilizing agent under these conditions. For protein-based stabilization using BSA, droplet stability was found to vary significantly with BSA concentration and the addition of dextrose. At the BSA concentrations studied, the droplets tended to coalesce shortly after formation, resulting in aggregation downstream from the microfluidic junction (i.e., the exit channel) (Figure 2B). This aggregation disrupted the flow, causing destabilization and backflow into the lateral flow channels and hindering consistent droplet production. However, when the BSA concentration was increased to 30 mg/mL with the addition of 50 mg/mL dextrose, a stable flow regime was achieved, allowing for the consistent generation of monodisperse

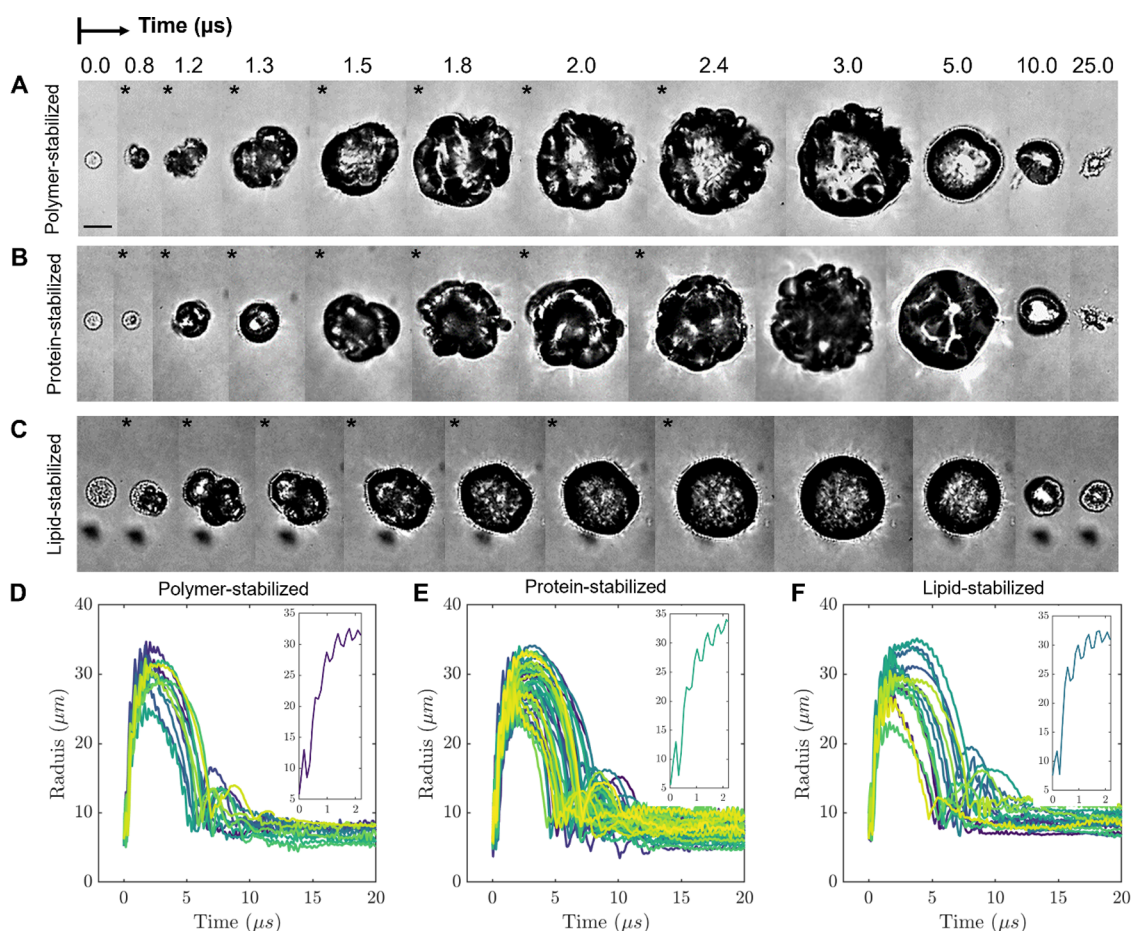


Figure 3. Ultrahigh-speed microscopy was used to quantify ADV bubble dynamics. Representative image sequences of (A) polymer-stabilized, (B) protein-stabilized, and (C) lipid-stabilized PFH droplets embedded in 10 mg/mL fibrin hydrogel, captured during and after ADV at 2.5 MHz (5 MPa peak negative pressure) are shown. Asterisks indicate presence of ultrasound pulse. Bubble radius versus time curves are shown for (D) polymer-stabilized ($n = 15$), (E) protein-stabilized ($n = 30$), and (F) lipid-stabilized ($n = 17$) droplets. Scale bar: 20 μm .

droplets (Figure 2C,D). Typically heating methods (e.g., sonication) are employed to denature albumin and achieve cross-linking. However, it is challenging to achieve elevated temperatures within a microfluidic device. Despite this, albumin-stabilized phase-shift droplets produced via microfluidics at room temperature displayed optimal storage stability, with no instances of spontaneous vaporization. Our findings are consistent with a previous study suggesting that stable microfluidic production of albumin-shelled microbubbles required albumin concentrations above 1% (w/v) and dextrose concentrations above 2.5% (w/v).³⁸ The stability of lipid-stabilized droplets was highly dependent on the inclusion of a PEG component in the formulation. Formulations lacking DPPE-PEG5k exhibited instability, often resulting in the formation of lipid films in the exit channel rather than discrete droplets (Figure 2E), indicating poor stabilization. The addition of DPPE-PEG5000 in the lipid formulation successfully stabilized the production of phase-shift droplets (Figure 2F). A 10% molar fraction of PEGylated lipid was selected, as previous research has shown this ratio to be optimal for achieving monodisperse, long-term stable conventional microbubbles.⁴⁷

The formulations that resulted in the stable production of monodisperse droplets (as shown in Figure 2A,D,F) were selected for further optical and acoustic characterization studies. To evaluate the effects of shell composition on droplet

storage stability, we monitored the average number-weighted diameter of polymer-, protein-, and lipid-stabilized droplets immediately following production and over a 14-day storage period (Figure 2G).

There were no significant changes in the average droplet diameter over the storage period for any of the three formulations, indicating excellent storage stability. Despite using identical flow rates during droplet production, lipid-stabilized droplets had significantly larger diameters compared to protein- and polymer-stabilized droplets. Droplet concentration was also assessed for each shell type. Polymer-stabilized droplets had the highest concentration at $(4.1 \pm 0.4) \times 10^8$ droplets/mL, followed by protein-stabilized droplets at $(3.2 \pm 0.8) \times 10^8$ droplets/mL, and lipid-stabilized droplets at $(1.9 \pm 0.4) \times 10^8$ droplets/mL. Over the 14-day storage period, droplet concentration remained stable across all formulations. The payload encapsulation efficiency was measured as $(98 \pm 2.6)\%$ for polymer-stabilized droplets and $(83 \pm 5.0)\%$ for lipid-stabilized droplets. Encapsulation efficiency for protein-stabilized droplets was not reported due to the autofluorescence of BSA within a similar emission spectrum,⁴⁸ which interfered with accurate quantification. Additionally, surface tension measurements yielded values of 71.1 ± 2.6 , 61.3 ± 4.0 , 50.0 ± 2.0 , and 47.4 ± 4.2 (mN/m) for DI water, protein (30 mg/mL BSA with 50 mg/mL dextrose), polymer (50 mg/mL Pluronic F68), and lipid (DPPC + DPPA + PEG5k) solutions,

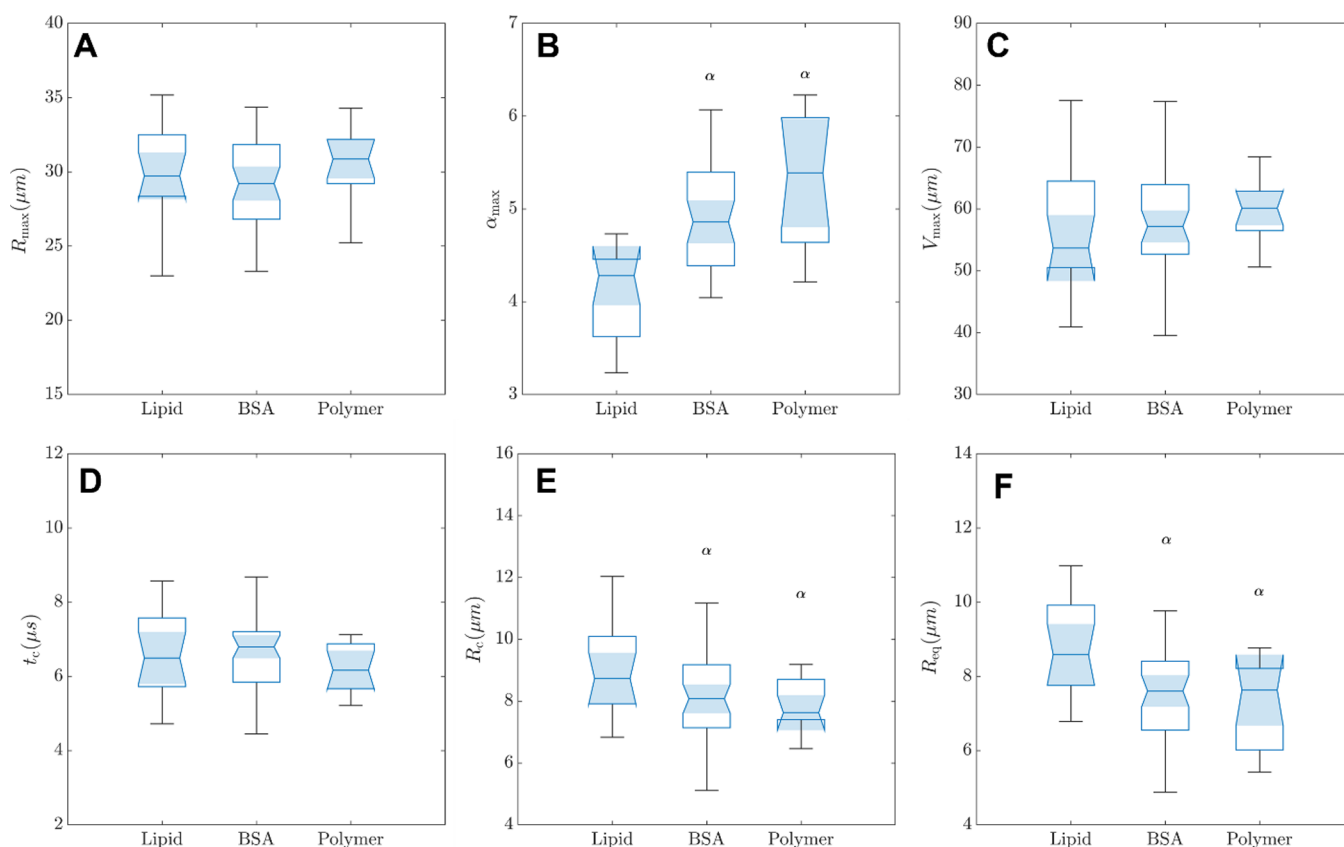


Figure 4. Ultrahigh-speed dynamics of acoustic droplet vaporization of PFH phase-shift droplets embedded in fibrin-based hydrogels were compared for three different shell materials. (A) Maximum radius ($R_{\max}$), (B) maximum expansion ratio ($\alpha_{\max}$), (C) maximum expansion velocity ($V_{\max}$), (D) collapse time (t_c), (E) collapse radius (R_c), and (F) quasi-equilibrium radius (R_{eq}) for each shell type is shown. Acoustic parameters were as follows: a single burst of 2.5 MHz, 5 MPa peak rarefactional pressure, and 6 cycles. Statistically significant differences ($p < 0.05$) are denoted as follows: α : vs lipid-stabilized droplets.

respectively (Figure 2H). The interfacial tension values between PFH and the shell solutions were 38.1 ± 5.1 , 33.9 ± 7.2 , and 35.3 ± 3.7 (mN/m) for protein, polymer, and lipid solutions, respectively.

Effect of Shell Composition on ADV Bubble Dynamics and Growth. To investigate the effect of shell composition on the dynamics of ADV-generated bubbles during and after ultrasound, two types of optical characterization techniques were employed: ultrahigh-speed brightfield imaging at 10 Mfps and time-lapse fluorescence confocal microscopy at 1 fps. Representative ultrahigh-speed image sequences of a polymer-, protein-, and lipid-stabilized PFH droplets embedded in 10 mg/mL fibrin hydrogel are shown in Figure 3, capturing bubble formation and evolution due to ADV. A single burst of ultrasound (2.5 MHz, 6 cycles, and 5 MPa peak rarefactional pressure) initiated ADV in less than a microsecond after ultrasound arrival. The rapid increase in bubble radius due to the phase-change as well as acoustically driven diffusion resulted in significant changes in radius that occurred on a microsecond time scale, corresponding to the pulse length of the acoustic wave. The expansion and contraction of the generated bubbles can be observed within the $2.4 \mu\text{s}$ burst of ultrasound (insets of Figure 3D–F). The radius-time curves show periodic spacing between peaks of approximately 400 ns, as shown in insets of Figure 3D–F, consistent with the period of the ultrasound pulse.

It should be noted that the duration of one complete bubble oscillation may exceed the ultrasound pulse period, depending

on factors such as bubble diameter, ultrasound frequency, and applied pressure. Radial oscillations (i.e., radius-time curves) were derived from the captured image sequences for each droplet (Figure 3D–F), allowing for the extraction of key kinematic parameters that describe bubble dynamics.

As the bubble expanded to its maximum radius during ultrasound ($R_{\max}$), it induced a local imbalance within the surrounding fibrin hydrogel, with the surrounding matrix experiencing stress greater than the internal pressure of the bubble. After reaching this peak radial expansion, each bubble contracted to a smaller radius (R_c) at a collapse time t_c , followed by multiple rebounds before stabilizing at a quasi-equilibrium radius measured 20 μs post ADV (R_{eq}). The entire imaging window, recorded at 10 Mfps, spanned 20 μs , during which the rapid dynamics of bubble expansion, collapse, rebound, and quasi stable radii were captured. Prior studies on shell elasticity (ξ_{shell}) and dilatational viscosity (κ_{shell}) have shown significant differences between lipid- and protein-shelled microbubbles. For example, Definity (ξ_{shell} : 0.8 N/m and κ_{shell} : 4×10^{-9} kg/s) and SonoVue (ξ_{shell} : 0.55 N/m and κ_{shell} : 7×10^{-9} kg/s) microbubbles, with a phospholipid shell, exhibit lower values compared to albumin-shelled Optison microbubbles (ξ_{shell} : 4 N/m and κ_{shell} : 3.2×10^{-8} kg/s).^{5,49,50} Polymer shells undergo complex changes in response to ultrasound, and models for their behavior are under development.⁵¹ In our high-speed studies, lipid-coated droplets exhibited a lower maximum radial expansion ratio ($\alpha_{\max}$: $R_{\max}/R_0$) compared to protein- and polymer-coated droplets,

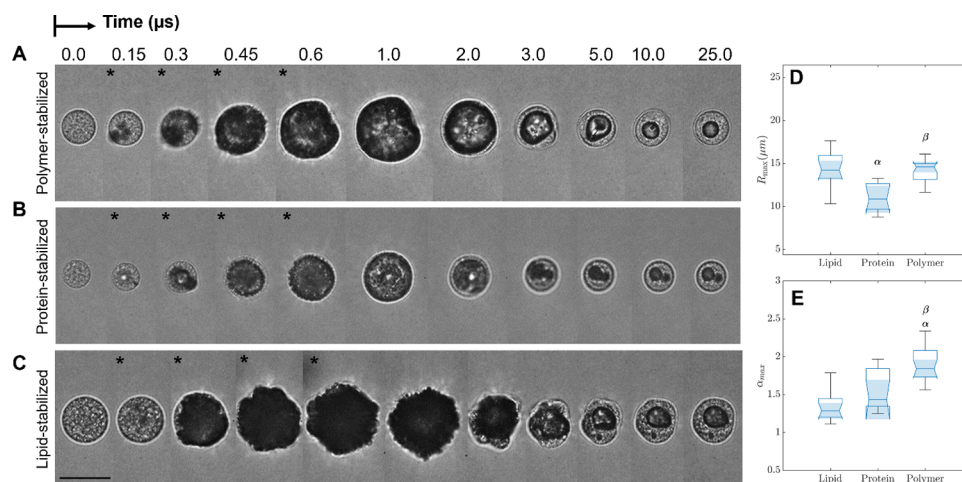


Figure 5. Ultrahigh-speed dynamics of phase-shift droplets at 9.6 MHz. Representative image sequences of (A) polymer-stabilized, (B) protein-stabilized, and (C) lipid-stabilized PFH droplets embedded in 10 mg/mL fibrin hydrogel, captured during and after ADV are shown. Asterisks denotes the presence of ultrasound. Box plots showing the maximum (D) expansion radius (R_{max}) and (E) expansion ratio (α_{max}) for lipid ($n = 14$), protein ($n = 10$), and polymer ($n = 20$) shells. Acoustic parameters were as follows: a single burst of 0.6 μs at 9.6 MHz (peak negative pressure: 5 MPa). Statistically significant differences ($p < 0.05$) are denoted as follows: α : vs lipid-stabilized droplets; β : vs protein-stabilized droplets. Scale bar: 15 μm .

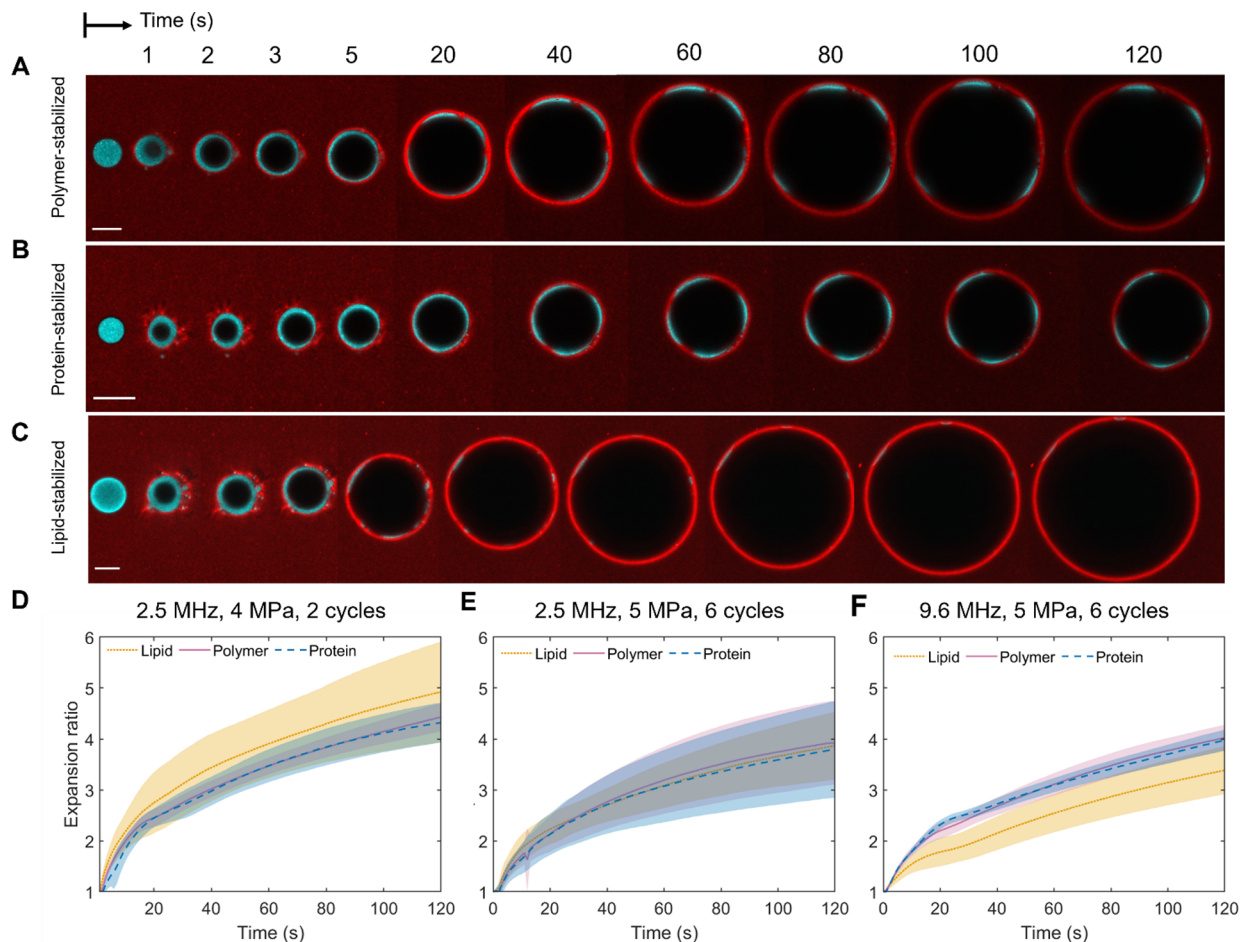


Figure 6. Real-time quasi-static growth of ADV-generated bubbles recorded using time-lapse confocal microscopy. Phase-shift droplets with (A) polymer-, (B) protein-, and (C) lipid-stabilized shells were embedded in fibrin-based hydrogels. The droplets (cyan) contained Cascade blue labeled dextran and the fibrin matrix (red) was labeled with Alexa Fluor 647 labeled fibrinogen. A single ultrasound burst induced droplet vaporization and payload release. Expansion ratio plots are shown for lipid-, polymer-, and protein-stabilized droplets at three different acoustic settings: (D) 2.5 MHz, 4 MPa, 2 cycles; (E) 2.5 MHz, 5 MPa, 6 cycles; and (F) 9.6 MHz, 5 MPa, 6 cycles. Shaded regions indicate the standard deviation ($n = 9-15$ for each shell and each condition). Scale bar: 20 μm .

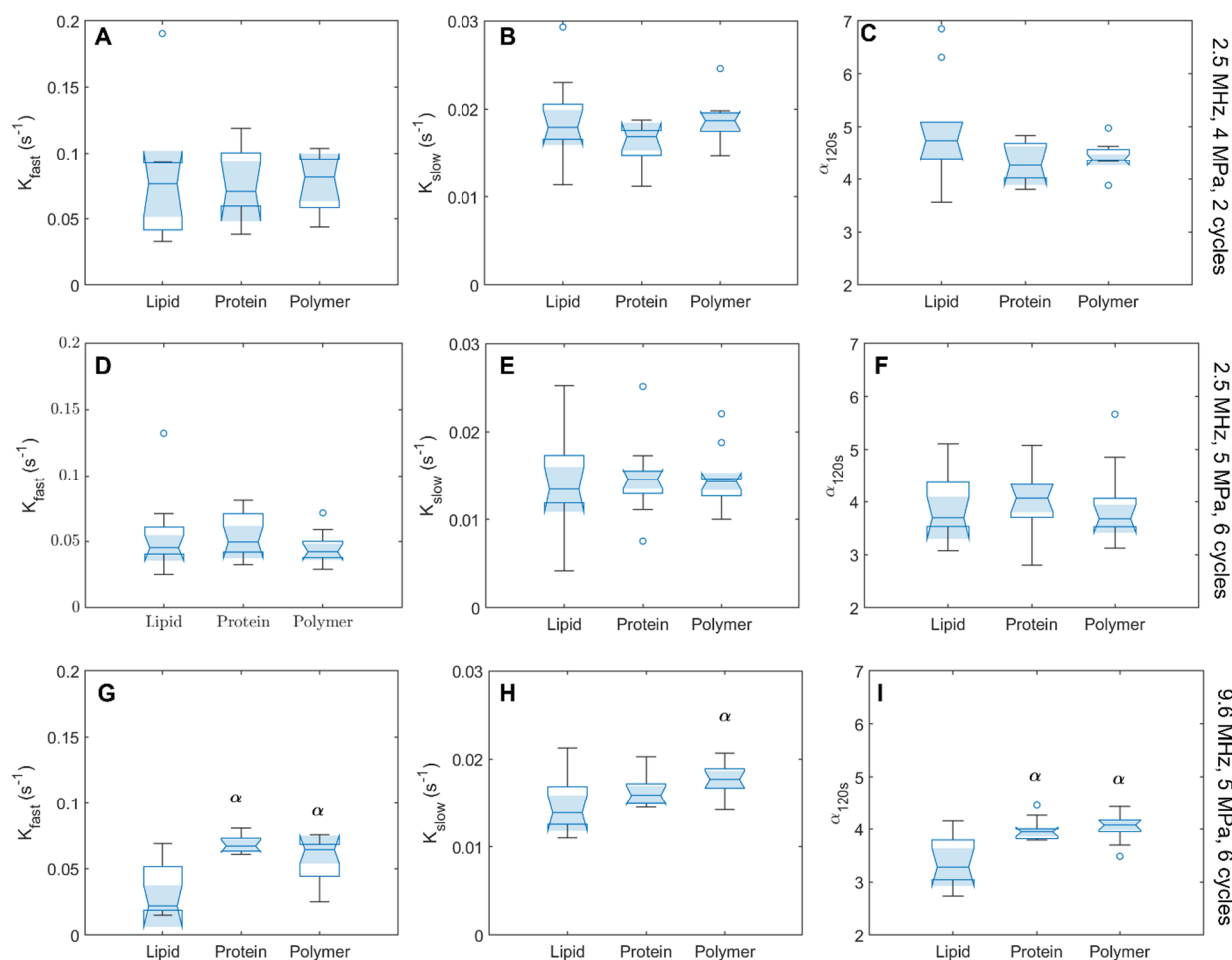


Figure 7. Growth rate constants and final expansion ratio of ADV-generated bubbles with different shell compositions. Box plots of fast growth rate constant (K_{fast}), slow growth rate constant (K_{slow}), and final expansion ratio at 120 s ($\alpha_{120\text{s}}$) for different shell materials were quantified using two-phase association model: (A–C) 2.5 MHz, 4 MPa, 2 cycles; (D–F) 2.5 MHz, 5 MPa, 6 cycles; and (G–I) 9.6 MHz, 5 MPa, 6 cycles. Statistically significant differences ($p < 0.05$) are denoted as follows: α : vs lipid-stabilized droplets.

likely due to the lipid shell's resistance to deformation. However, despite their smaller initial expansion, lipid-stabilized droplets attained larger collapse (R_c) and quasi-equilibrium (R_{eq}) radii compared to the other shell types. The collapse time and expansion velocity did not differ significantly among the three shell compositions.

Accurately measuring the collapse time and radius of ADV bubbles is challenging due to the rapid collapse dynamics and difficulty in capturing the exact moment when the bubble reaches its absolute minimum size. At 9.6 MHz (6 cycles, 5 MPa), all three shell compositions displayed significantly smaller radial expansions compared to 2.5 MHz (Figure 5). Similar to the results observed at 2.5 MHz (Figure 4B), lipid-stabilized droplets underwent smaller expansion during ADV at 9.6 MHz as well (Figure 5E). Since ADV was more localized and confined within the phase-shift droplet at 9.6 MHz, accurately measuring R_c and t_c was challenging. Consequently, only R_{max} and α_{max} were measured under these conditions. Knowing R_{eq} and R_{max} can help in biomedical applications such as sonoporation, where the distance of cells with respect to phase-shift droplets (d) can define cellular bioeffects such as irreversible sonoporation ($d < R_{\text{eq}}$), reversible sonoporation ($R_{\text{eq}} < d < R_{\text{max}}$), or no sonoporation ($d > R_{\text{max}}$).^{52–54}

To track the quasi-static growth of the ADV-generated bubbles—where gas diffusion drives the expansion on a much

longer time scale—confocal imaging was performed at 1 fps over a 120-s period. Confocal microscopy images show phase-shift droplets with a fluorescently labeled liquid core (displayed in cyan) embedded within a fibrin matrix (in red) for a polymer- (Figure 6A), protein- (Figure 6B), and lipid-stabilized (Figure 6C) droplets. Prior to ADV, the droplets effectively retained the fluorescent dye (Cascade Blue dextran) within their cores, indicating effective payload encapsulation. Upon exposure to a single ultrasound burst (2.5 MHz, 5 MPa, and 6 cycles), vaporization was triggered, resulting in release of the payload. This was evident by the absence of fluorescence within the newly formed bubbles. Although aspherical bubble morphologies were recorded during ultrasound (Figures 3 and 5), the generated bubbles stabilized into a spherical shape on a longer time scale (Figure 6) and did not undergo fragmentation or rapid dissolution during the observation period. Instead, they exhibited a gradual, diffusion-driven increase in size, illustrated by the expansion ratio plots in Figures 6D–F. The prolonged stability can be attributed to the unique properties of the PFH core, which has low diffusivity and solubility in water. Additionally, the larger initial droplet diameter and the compressed fibrin hydrogel at the gas-fibrin interface likely contribute to the ADV-bubble stability under high pressures. This diffusion-driven growth, occurring over seconds, is notably slower—by 6 orders of magnitude—than

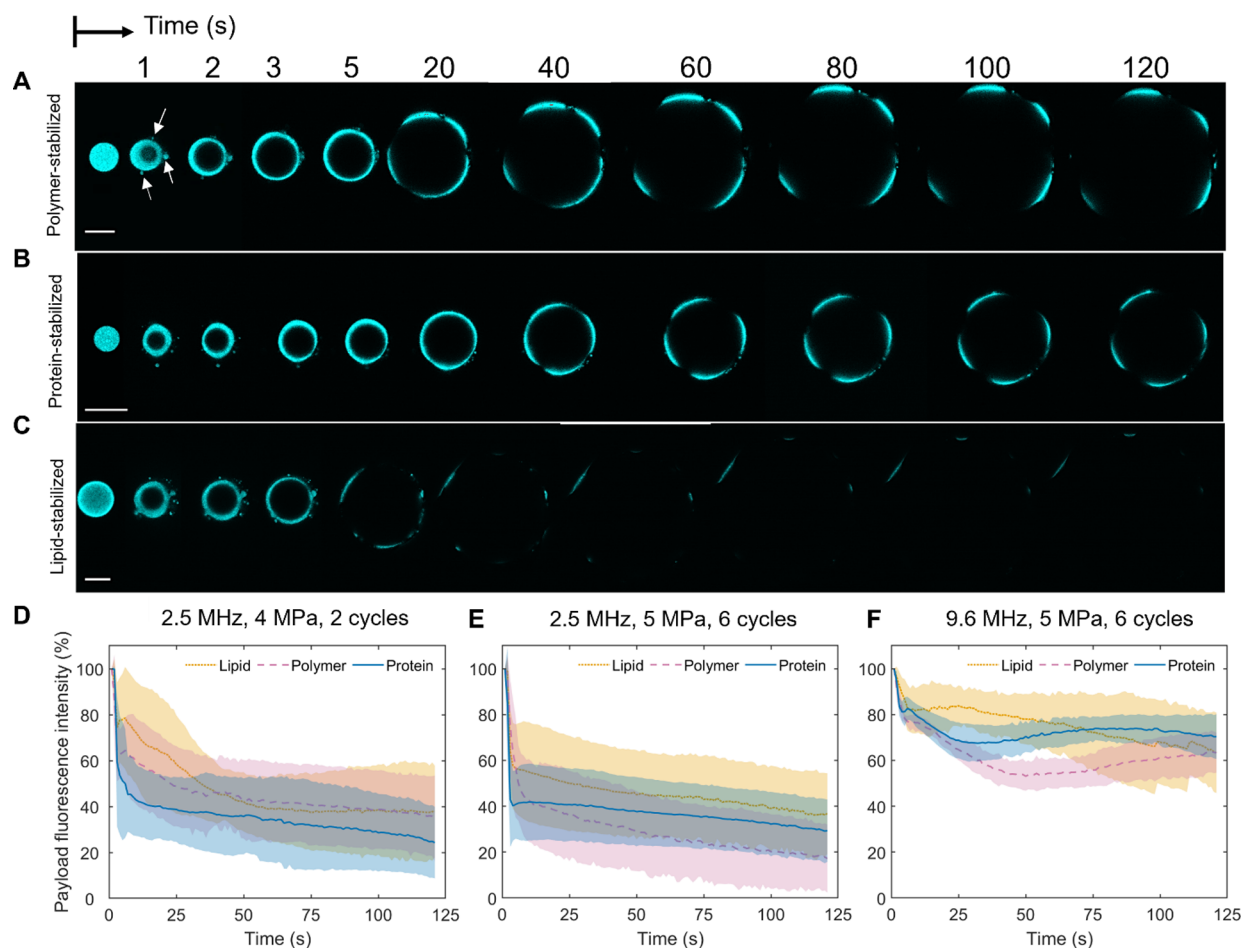


Figure 8. Real-time fluorescence tracking of payload release from phase-shift droplets post-ADV. Time-lapse confocal microscopy images of payload-carrying droplets with (A) polymer, (B) protein, and (C) lipid shells are shown at the following acoustic settings: 2.5 MHz, 5 MPa, 6 cycles. The droplets (blue) contain Cascade Blue labeled dextran. Arrows indicate the expelled payloads during ADV. Longitudinal fluorescence decay curves are shown for each shell composition under different acoustic parameters: (D) 2.5 MHz, 4 MPa, 2 cycles; (E) 2.5 MHz, 5 MPa, 6 cycles; and (F) 9.6 MHz, 5 MPa, 6 cycles (Shaded regions indicate the standard deviation, $n = 9–15$ for each droplet and each condition). Scale bar: 20 μm .

the rapid ADV dynamics seen on the microsecond time scale (Figure 3).

At 2.5 MHz, two acoustic settings were interrogated: 4 MPa (2 cycles) and 5 MPa (6 cycles). The temporal evolution of the ADV-generated bubble growth under these conditions, as shown in Figure 6D–F, was further analyzed using a two-phase association model.

This model quantified parameters such as the fast growth rate constant (K_{fast}), slow growth rate constant (K_{slow}), and the final bubble expansion ratio at 120 s ($\alpha_{120\text{s}}$). Results at 2.5 MHz showed no statistically significant differences in growth behavior among the shell materials, indicating comparable growth characteristics across these shell types under these conditions (Figure 7A–F). However, at 9.6 MHz, ADV-generated bubbles from lipid-coated droplets displayed significantly lower values for K_{fast} , K_{slow} , and $\alpha_{120\text{s}}$ compared to other shell types (Figure 7G–I). This slower growth may be attributed to the intact lipid shells following ADV, which could limit gas diffusion and constrain bubble growth. Conversely, protein and polymer shells likely experienced rupture or cracking during ADV even at low expansion ratios induced at 9.6 MHz, resulting in incomplete and uneven coverage of the generated bubble. This lack of homogeneous coverage would facilitate faster gas diffusion, thereby enabling more rapid

bubble growth and payload release. This finding is consistent with a previous study that compared the growth dynamics of lipid-, BSA-, and Krytox-stabilized perfluoropentane droplets over 15 min, with significantly faster growth up to 50 times the initial size for BSA- and Krytox-stabilized droplets, compared to 15 times for lipid-stabilized droplets at 10 MHz.⁴¹

The fluorescence intensity of the payload released by ADV was also tracked longitudinally over a period of 120-s period post-ADV using time-lapse confocal microscopy (Figure 8). Real-time confocal microscopy revealed a significant decrease in payload fluorescence intensity after ADV, with expelled payloads visible in the 1-s frame following vaporization for all the shell types. As shown in Figure 8D–E, the fluorescence intensity at 1 s post-ADV indicated that approximately (25 ± 10) , (35 ± 13) , and $(40 \pm 20)\%$ of the payload was released from lipid-, polymer-, and protein-stabilized droplets, respectively, during ADV at 2.5 MHz. The released payload was observed both in the form of diffused cloud release and expelled vesicles. Following ADV, the fluorescence marker exhibited an initially uniform distribution along the surface of the newly formed bubbles. Over time, however, the fluorescence pattern became increasingly heterogeneous, with distinct patches of brighter fluorescence and regions lacking visible fluorescence. This pattern suggests a dynamic

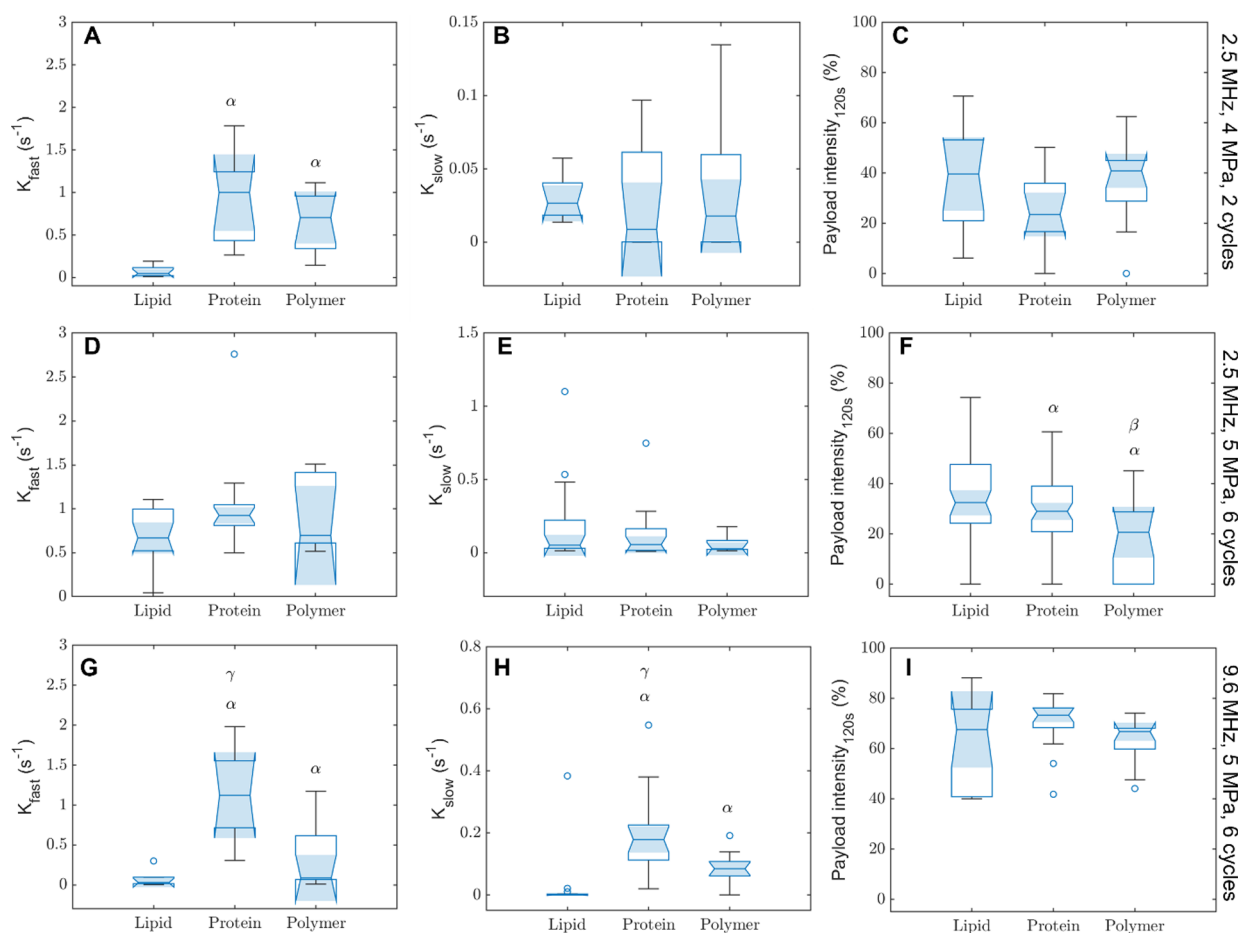


Figure 9. Quantification of payload release kinetics and retention for ADV-generated bubbles with different shell compositions. Box plots illustrating the fast release rate constant (K_{fast}), slow release rate constant (K_{slow}), and payload intensity at 120 s post-ADV for lipid-, protein-, and polymer-stabilized droplets under three different acoustic conditions: (A–C) 2.5 MHz, 4 MPa, 2 cycles; (D–F) 2.5 MHz, 5 MPa, 6 cycles; and (G–I) 9.6 MHz, 5 MPa, 6 cycles. Statistically significant differences ($p < 0.05$) are denoted as follows: α : vs lipid-stabilized droplets; β : vs protein-stabilized droplets; γ : vs polymer-stabilized droplets.

reorganization or redistribution of the payload at the bubble interface, possibly in response to changes in surface energy or the surrounding medium over time. Diffusion driven growth of the ADV-generated bubble continued release of the remaining payload, reaching (63 ± 17) , (75 ± 15) , and $(70 \pm 13)\%$ for lipid-, polymer-, and protein-stabilized droplets, respectively, after 120 s (Figure 9). At 9.6 MHz, the initial payload released during ADV was only up to 15% (1-s frame), with subsequent bubble growth further increasing release up to 35% after 120 s.

The fluorescence decay profiles, representing the reduction in payload fluorescence intensity over time, were fitted to an exponential decay model (Figure 9). This model was used to quantify the rate of payload release and to calculate the residual payload retention at 120 s post-ADV. K_{fast} was significantly higher for protein- and polymer-stabilized droplets compared to lipid-stabilized droplets (Figure 9A,G). A comparison of bubble growth rate constants in Figure 7 with payload release rate constants in Figure 9 reveals that the payload release rate was approximately an order of magnitude faster than the bubble growth rate for PFH phase-shift droplets.

In our optical studies, we observed stable bubble formation (without recondensation) for all phase-shift droplets, regardless of shell composition, using a single burst of ultrasound. Although we did not measure ADV efficiency, which is typically assessed at higher volume fractions relevant to

biomedical applications, we did not observe stochastic vaporization for shell-stabilized droplets under our experimental conditions. Prior studies have shown that while single surfactant-shell microdroplets and double-shell microdroplets (surfactant–polymer layers) exhibited similar expansion ratios during ADV, microdroplets with a more rigid polymeric shell achieved the highest ADV efficiency. This increased efficiency was attributed to the thick polymer layer, which reduced gaseous perfluorocarbon leakage and enhanced overall stability.³³

Phase-Shift Droplets with Fluorescently Labeled Shells. Three-dimensional (3D) maximum intensity projection confocal images confirmed that PFH droplets were encapsulated by fluorescently labeled shells of BSA (Figure 10A) and lipid (Figure 10B) prior to ADV. Following ADV, a 3D image captured at 60 s (Figure 10C) displays the generated bubble from a lipid-stabilized droplet, illustrating the postvaporization state. Sequential two-dimensional (2D) images of the fluorescently labeled payload (cyan) and shell (green) revealed that shell thinning corresponded with payload release. Regions where the shell fluorescence signal persisted were associated with payload retention (Figure 10D,E). Fluorescence intensity analysis revealed that photobleaching accounted for up to a 15% decrease in shell fluorescence for both lipid- and protein-stabilized droplets over time (Figure

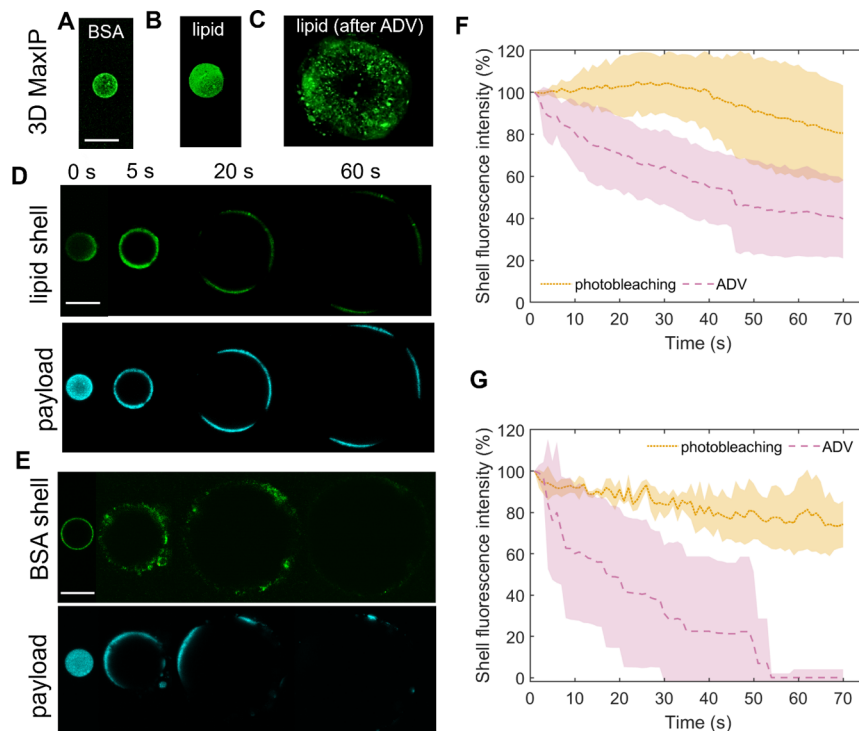


Figure 10. Real-time tracking of the fluorescently labeled lipid and BSA shells in phase-shift droplets. 3D maximum intensity projection images showing complete encapsulation of PFH droplets by fluorescently labeled (A) BSA and (B) lipid shells before ADV, and (C) the postvaporization state of a lipid-stabilized bubble at 60 s after ADV. 2D time-lapse images of the fluorescently labeled payload (cyan), and shell (green) showing fluorescence distribution over time post-ADV are presented for (D) lipid and (E) BSA. Shell fluorescence intensity decay over time for (F) lipid- ($n = 12$) and (G) protein-stabilized ($n = 9$) droplets, with a comparison between photobleaching and ADV-induced fluorescence loss (Shaded regions indicate the standard deviation). Acoustic parameters were as follows: a single burst of 2.5 MHz, 5 MPa peak rarefactional pressure, and 6 cycles. Scale bar: 15 μm .

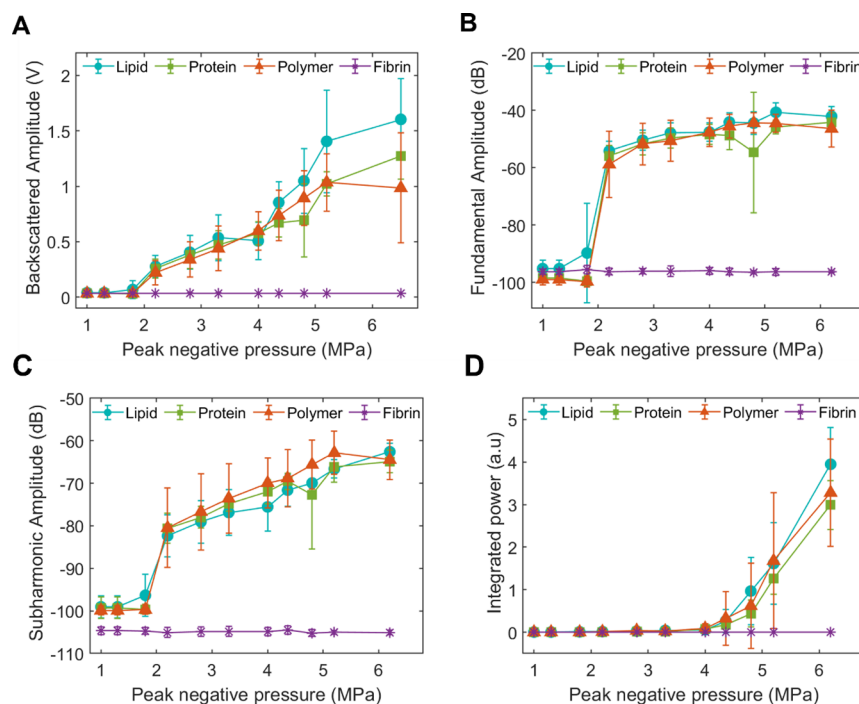


Figure 11. Acoustic emissions from ADV-generated bubbles with different shell compositions. (A) Backscattered amplitude of ADV-generated bubbles as a function of peak negative pressure, used to determine the ADV threshold for lipid-, protein-, and polymer-stabilized droplets. (B) Fundamental and (C) subharmonic amplitudes of the acoustic emissions, demonstrating significant increases due to ADV compared to fibrin-only hydrogels. (D) Integrated power frequency spectrum was plotted as a function of peak negative pressure to measure the inertial cavitation threshold. Acoustic parameters were: 2.5 MHz, 3 μs pulse duration, and pulse repetition period of 100 Hz. $n = 6$ for each shell material.

10F,G). However, ADV-induced vaporization resulted in a more substantial reduction in shell fluorescence, with approximately a 60% decrease observed in lipid-stabilized droplets (Figure 10F) and nearly a 100% decrease in protein-stabilized droplets (Figure 10G), indicating substantial shell disruption post-ADV.

Effect of Shell Composition on the Acoustic Emissions. The acoustic emissions from ADV-generated bubbles can be utilized for a wide range of applications, including ultrasound localization microscopy for super-resolution imaging,²⁹ image-guided cancer therapy,⁵⁵ and tissue characterization.⁴⁶ We characterized the acoustic emissions from ADV-generated bubbles stabilized with lipid, protein, and polymer shells using a passive cavitation detection technique to assess key parameters such as the ADV threshold, inertial cavitation threshold, fundamental, and subharmonic amplitudes. The pressure-dependent backscattered amplitude of the generated bubbles, as shown in Figure 11A, was analyzed by a piece-wise linear function to estimate the ADV threshold for each shell composition. The ADV threshold for lipid-, protein-, and polymer-stabilized droplets was 1.8 ± 0.05 , 1.8 ± 0.02 , and 1.8 ± 0.1 MPa, respectively. The results suggest that the shell composition did not significantly impact the ADV threshold. Further analysis of the acoustic emissions quantified the fundamental (Figure 11B) and subharmonic (Figure 11C) scattered amplitudes. ADV-generated bubbles demonstrated significant increases in both fundamental and subharmonic responses, reaching up to 50 and 45 dB, respectively, compared to fibrin-only hydrogels. The scattered fundamental response from PFH droplets saturated at excitation pressures above 2.4 MPa for the three different shell compositions. The integrated power of the acoustic emissions as a function of peak negative pressure was also analyzed to determine the inertial cavitation threshold of ADV-bubbles (Figure 11D). The inertial cavitation threshold was 4.4 ± 0.4 , 4.5 ± 0.2 , and 4.6 ± 0.6 MPa for lipid-, protein-, and polymer-stabilized droplets, respectively. No inertial cavitation threshold was detected for fibrin-only hydrogels, confirming that the observed acoustic responses were specific to the phase-shift droplets.

Although our high-speed imaging results indicated a larger maximum expansion ratio for protein- and polymer-stabilized droplets (Figures 4B and 5E), shell material did not significantly affect the acoustic response of bubbles generated from phase-shift droplets. It should be noted that optical interrogations were conducted at a much lower droplet volume fraction (i.e., 0.006%) and with a single burst of $2.4 \mu\text{s}$ at 2.5 MHz to better understand the underlying physics, while acoustic studies were performed at a higher droplet concentration (i.e., 0.05%) using clinically relevant acoustic parameters, such as short bursts of $3 \mu\text{s}$ at a repetition rate of 100 Hz and a frequency of 2.5 MHz (data are shown as mean $\pm$ standard deviation).

While a previous study suggested that the energy required to thermally vaporize nanodroplets increased with longer acyl chain lengths (from C14 to C20),⁵⁶ we observed no notable differences in the ADV threshold across the different shell compositions. Prior numerical studies have suggested that the ADV threshold could depend on the shear modulus of the shell, particularly for submicron-sized droplets with shear modulus >400 MPa.⁵⁷ These models assume the shell remains intact and homogeneously covers the bubble/droplet system during ADV. Our real-time confocal imaging showed that the

shell did not remain homogeneous post-ADV, as rupture or significant deformation occurred. The initiation of ADV is generally believed to occur through homogeneous nucleation within the droplet, as demonstrated by several high-speed imaging studies,^{21,58,59} indicating that the ADV threshold should be independent of shell composition. However, we initially anticipated that shell material might impact the acoustic response of the ADV-generated bubbles, as prior research has shown that the nonlinear behavior and destruction thresholds of conventional microbubbles are influenced by shell composition.^{5,18} It should be noted that encapsulated microbubbles exhibit a nonlinear response at low acoustic pressure amplitudes (below 100 kPa), with destruction and fragmentation occurring at 100–150 kPa for soft-shelled microbubbles and above 300 kPa for hard-shelled microbubbles.⁵ In the case of ADV, which occurs at peak negative pressures in the megapascal range, the effect of the initial shell material on radial oscillation appears to become negligible. Studies on conventional microbubbles, such as Optison (albumin-shelled with a C_3F_8 gas core) and MP1950 (lipid-shelled with a C_4F_{10} gas core), have similarly shown that oscillation amplitude and fragmentation under high acoustic pressures (>1.4 MPa) are largely independent of shell characteristics.⁶⁰

It should be noted that this study focused on the role of shell composition for micron-sized droplets. The influence of shell material on smaller-sized droplets, such as nanodroplets, remains to be investigated and may be more pronounced. The maximum expansion of phase-shift droplets is size-dependent, with smaller droplets exhibiting reduced expansion due to Laplace overpressure.²⁹ At smaller droplet sizes, where expansion is limited, the shell's effect may follow trends observed in conventional microbubbles. Classical nucleation theory predicted a threshold radius of $5.2 \mu\text{m}$ for PFH bubbles, above which ADV-generated bubbles remained stable against recondensation.⁶¹ However, the effect of recondensation was not studied here. While lipid, protein, and polymer shells each have unique properties, our findings suggest that they can be tailored to meet specific therapeutic requirements without significantly impacting the ADV and inertial cavitation thresholds. This adaptability highlights the potential of phase-shift droplets as versatile, customizable platforms for targeted drug delivery and diagnostic imaging applications.

CONCLUSIONS

We investigated the role of the encapsulating shell composition on the stability, radial dynamics, payload release, and acoustic emissions of phase-shift droplets. Monodisperse droplets with three commonly used shell materials, lipid, protein, or polymer were successfully generated using a microfluidic technique, with conditions and formulations optimized to ensure stable flow during droplet production. For example, albumin-stabilized droplets required a concentration of 30 mg/mL BSA and 50 mg/mL dextrose for consistent monodisperse production, while lipid-stabilized droplets relied on the inclusion of 10% PEGylated lipids to achieve stability. Over a 14-day storage period, droplets of all three shell types exhibited no significant changes in size or concentration, demonstrating excellent long-term stability. Optical characterization through high-speed imaging and time-lapse confocal microscopy provided crucial insights into the radial dynamics and growth of droplets with varying shell materials, indicating that shell composition may have a more pronounced influence

at higher excitation frequencies where radial expansion is reduced. Fluorescent labeling allowed us to quantify payload release and shell retention, highlighting the distinct properties of each shell type. The heterogeneous patterns observed in the shell fluorescence signal suggested partial rupture or nonuniform encapsulation of the generated bubbles. While microbubble acoustic responses are known to depend on shell properties, the influence of the shell on phase-shift droplets becomes negligible due to the volumetric expansion during ADV, which governs their acoustic behavior. These findings emphasize the versatility of phase-shift droplets as tunable platforms for a variety of diagnostic and therapeutic applications.

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Notes

The authors declare no competing financial interest.

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REFERENCES

- (1) Kripfgans, O. D.; Fowlkes, J. B.; Miller, D. L.; Eldevik, O. P.; Carson, P. L. Acoustic droplet vaporization for therapeutic and diagnostic applications. *Ultrasound Med. Biol.* **2000**, *26* (7), 1177–1189.
- (2) Aliabouzar, M.; Kripfgans, O. D.; Fowlkes, J. B.; Fabiilli, M. L. Bubble nucleation and dynamics in acoustic droplet vaporization: a review of concepts, applications, and new directions. *Z. Med. Phys.* **2023**, *33* (3), 387–406.
- (3) Lea-Banks, H.; O'reilly, M.; Hynynen, K. Ultrasound-responsive droplets for therapy: A review. *J. Controlled Release* **2019**, *293*, 144–154.
- (4) Ferrara, K.; Pollard, R.; Borden, M. Ultrasound microbubble contrast agents: fundamentals and application to gene and drug delivery. *Annu. Rev. Biomed. Eng.* **2007**, *9* (1), 415–447.
- (5) Frinking, P.; Segers, T.; Luan, Y.; Tranquart, F. Three decades of ultrasound contrast agents: a review of the past, present and future improvements. *Ultrasound Med. Biol.* **2020**, *46* (4), 892–908.
- (6) Church, C. C. The effects of an elastic solid surface layer on the radial pulsations of gas bubbles. *J. Acoust. Soc. Am.* **1995**, *97* (3), 1510–1521.
- (7) Sarkar, K.; Shi, W. T.; Chatterjee, D.; Forsberg, F. Characterization of ultrasound contrast microbubbles using in vitro experiments and viscous and viscoelastic interface models for encapsulation. *J. Acoust. Soc. Am.* **2005**, *118* (1), 539–550.
- (8) Sirsi, S.; Borden, M. Microbubble compositions, properties and biomedical applications. *Bubble Sci., Eng., Technol.* **2009**, *1* (1–2), 3–17.
- (9) Christiansen, C.; Skotland, T. Changes of protein solutions during storage: a study of albumin pharmaceutical preparations. *Biotechnol. Appl. Biochem.* **2010**, *55* (3), 121–130.
- (10) Pisani, E.; Tsapis, N.; Paris, J.; Nicolas, V.; Cattel, L.; Fattal, E. Polymeric nano/microcapsules of liquid perfluorocarbons for ultrasonic imaging: physical characterization. *Langmuir* **2006**, *22* (9), 4397–4402.
- (11) Kooiman, K.; Vos, H. J.; Versluis, M.; De Jong, N. Acoustic behavior of microbubbles and implications for drug delivery. *Adv. Drug Delivery Rev.* **2014**, *72*, 28–48.
- (12) Frinking, P. J.; De Jong, N. Acoustic modeling of shell-encapsulated gas bubbles. *Ultrasound Med. Biol.* **1998**, *24* (4), 523–533.
- (13) Sojahrood, A.; Haghi, H.; Porter, T.; Karshafian, R.; Kolios, M. Experimental and numerical evidence of intensified non-linearity at the microscale: The lipid coated acoustic bubble. *Phys. Fluids* **2021**, *33* (7), No. 072006.
- (14) Roovers, S.; Segers, T.; Lajoinie, G.; Deprez, J.; Versluis, M.; De Smedt, S. C.; Lentacker, I. The role of ultrasound-driven microbubble dynamics in drug delivery: from microbubble fundamentals to clinical translation. *Langmuir* **2019**, *35* (31), 10173–10191.
- (15) Frinking, P. J.; de Jong, N.; Cespedes, E. I. Scattering properties of encapsulated gas bubbles at high ultrasound pressures. *J. Acoust. Soc. Am.* **1999**, *105* (3), 1989–1996.
- (16) Lee, S.; Kim, D. H.; Needham, D. Equilibrium and dynamic interfacial tension measurements at microscopic interfaces using a micropipet technique. 2. Dynamics of phospholipid monolayer formation and equilibrium tensions at the water–air interface. *Langmuir* **2001**, *17* (18), 5544–5550.
- (17) Dayton, P. A.; Morgan, K. E.; Klivanov, A. L.; Brandenburger, G.; Ferrara, K. W. Optical and acoustical observations of the effects of ultrasound on contrast agents. *IEEE Trans. Sonics Ultrason.* **1999**, *46* (1), 220–232.
- (18) Borden, M. A.; Kruse, D. E.; Caskey, C. F.; Zhao, S.; Dayton, P. A.; Ferrara, K. W. Influence of lipid shell physicochemical properties on ultrasound-induced microbubble destruction. *IEEE Trans. Sonics Ultrason.* **2005**, *52* (11), 1992–2002.
- (19) Qamar, A.; Wong, Z. Z.; Fowlkes, J. B.; Bull, J. L. Dynamics of acoustic droplet vaporization in gas embolotherapy. *Appl. Phys. Lett.* **2010**, *96* (14), 143702.
- (20) Abeid, B. A.; Fabiilli, M. L.; Estrada, J. B.; Aliabouzar, M. Ultra-high-speed dynamics of acoustic droplet vaporization in soft biomaterials: Effects of viscoelasticity, frequency, and bulk boiling point. *Ultrason. Sonochem.* **2024**, *103*, No. 106754.
- (21) Kripfgans, O. D.; Fabiilli, M. L.; Carson, P. L.; Fowlkes, J. B. On the acoustic vaporization of micrometer-sized droplets. *J. Acoust. Soc. Am.* **2004**, *116* (1), 272–281.

- (22) Spiekhou, S.; van Elburg, B.; Voorneveld, J.; Jong, N. d.; Versluis, M.; Bosch, J. G.; Segers, T. Are monodisperse phospholipid-coated microbubbles “mono-acoustic”? *Appl. Phys. Lett.* **2024**, *124*, 231601 DOI: 10.1063/5.0215736.
- (23) Segers, T.; Kruizinga, P.; Kok, M. P.; Lajoinie, G.; De Jong, N.; Versluis, M. Monodisperse versus polydisperse ultrasound contrast agents: non-linear response, sensitivity, and deep tissue imaging potential. *Ultrasound Med. Biol.* **2018**, *44* (7), 1482–1492.
- (24) van Elburg, B.; Deprez, J.; van den Broek, M.; De Smedt, S. C.; Versluis, M.; Lajoinie, G.; Lentacker, I.; Segers, T. Dependence of sonoporation efficiency on microbubble size: An in vitro monodisperse microbubble study. *J. Controlled Release* **2023**, *363*, 747–755.
- (25) Fabiilli, M. L.; Lee, J. A.; Kripfgans, O. D.; Carson, P. L.; Fowlkes, J. B. Delivery of water-soluble drugs using acoustically triggered perfluorocarbon double emulsions. *Pharm. Res.* **2010**, *27*, 2753–2765.
- (26) Couture, O.; Faivre, M.; Pannacci, N.; Babataheri, A.; Servois, V.; Tabeling, P.; Tanter, M. Ultrasound internal tattooing. *Medical physics* **2011**, *38* (2), 1116–1123.
- (27) Benton, R. P.; Al Rifai, N.; Stone, K.; Clark, A.; Zhang, B.; Haworth, K. J. Impact of Perfluoropentane Microdroplets Diameter and Concentration on Acoustic Droplet Vaporization Transition Efficiency and Oxygen Scavenging. *Pharmaceutics* **2022**, *14* (11), 2392.
- (28) Melich, R.; Bussat, P.; Morici, L.; Vivien, A.; Gaud, E.; Bettinger, T.; Cherkaoui, S. Microfluidic preparation of various perfluorocarbon nanodroplets: Characterization and determination of acoustic droplet vaporization (ADV) threshold. *Int. J. Pharm.* **2020**, *587*, No. 119651.
- (29) Burgess, M. T.; Aliabouzar, M.; Aguilar, C.; Fabiilli, M. L.; Ketterling, J. A. Slow-flow ultrasound localization microscopy using recondensation of perfluoropentane nanodroplets. *Ultrasound Med. Biol.* **2022**, *48* (5), 743–759.
- (30) Zhang, X.; Hu, J.; Zhao, G.; Huang, N.; Tan, Y.; Pi, L.; Huang, Q.; Wang, F.; Wang, Z.; Wang, Z. PEGylated PLGA-based phase shift nanodroplets combined with focused ultrasound for blood brain barrier opening in rats. *Oncotarget* **2017**, *8* (24), 38927.
- (31) Moncion, A.; Kripfgans, O. D.; Carson, P. L.; Fowlkes, J. B.; Fabiilli, M. L. Characterization of acoustic droplet vaporization and inertial cavitation thresholds in acoustically-responsive tissue scaffolds. In *2014 IEEE International Ultrasonics Symposium*; IEEE, 2014; pp 1646–1649.
- (32) Capece, S.; Domenici, F.; Brasili, F.; Oddo, L.; Cerroni, B.; Bedini, A.; Bordini, F.; Chiessi, E.; Paradossi, G. Complex interfaces in “phase-change” contrast agents. *Phys. Chem. Chem. Phys.* **2016**, *18* (12), 8378–8388.
- (33) Palmieri, D.; Brasili, F.; Capocefalo, A.; Bizien, T.; Angelini, I.; Oddo, L.; Toumia, Y.; Paradossi, G.; Domenici, F. Improved hybrid-shelled perfluorocarbon microdroplets as ultrasound- and laser-activated phase-change platform. *Colloids Surf. Physicochem. Eng. Aspects* **2022**, *641*, No. 128522.
- (34) Garg, S.; Thomas, A. A.; Borden, M. A. The effect of lipid monolayer in-plane rigidity on in vivo microbubble circulation persistence. *Biomaterials* **2013**, *34* (28), 6862–6870.
- (35) Xiao, H.; Aliabouzar, M.; Fabiilli, M. L. Acoustically responsive scaffolds: Unraveling release kinetics and mechanisms for sustained, steady drug delivery. *J. Controlled Release* **2024**, *374*, 205–218.
- (36) Unger, E. C.; Porter, T.; Culp, W.; Labell, R.; Matsunaga, T.; Zutshi, R. Therapeutic applications of lipid-coated microbubbles. *Adv. Drug Delivery Rev.* **2004**, *56* (9), 1291–1314.
- (37) Kwan, J. J.; Borden, M. A. Lipid monolayer collapse and microbubble stability. *Adv. Colloid Interface Sci.* **2012**, *183*, 82–99.
- (38) Chen, J. L.; Dhanaliwala, A. H.; Dixon, A. J.; Klivanov, A. L.; Hossack, J. A. Synthesis and characterization of transiently stable albumin-coated microbubbles via a flow-focusing microfluidic device. *Ultrasound Med. Biol.* **2014**, *40* (2), 400–409.
- (39) Aliabouzar, M.; Ley, A. W.; Meurs, S.; Putnam, A. J.; Baker, B. M.; Kripfgans, O. D.; Fowlkes, J. B.; Fabiilli, M. L. Micropatterning of acoustic droplet vaporization in acoustically-responsive scaffolds using extrusion-based bioprinting. *Bioprinting* **2022**, *25*, No. e00188.
- (40) Bou, S.; Klymchenko, A. S.; Collot, M. Fluorescent labeling of biocompatible block copolymers: Synthetic strategies and applications in bioimaging. *Materials advances* **2021**, *2* (10), 3213–3233.
- (41) Reznik, N.; Seo, M.; Williams, R.; Bolewska-Pedyczak, E.; Lee, M.; Matsuura, N.; Gariepy, J.; Foster, F. S.; Burns, P. N. Optical studies of vaporization and stability of fluorescently labelled perfluorocarbon droplets. *Phys. Med. Biol.* **2012**, *57* (21), 7205.
- (42) Aliabouzar, M.; Jivani, A.; Lu, X.; Kripfgans, O. D.; Fowlkes, J. B.; Fabiilli, M. L. Standing wave-assisted acoustic droplet vaporization for single and dual payload release in acoustically-responsive scaffolds. *Ultrason. Sonochem.* **2020**, *66*, No. 105109.
- (43) Kirkkitadze, M.; Remi, E.; Bhandal, K.; Carpick, B. Visible and subvisible particles in the BCG immunotherapeutic product Immucyst®. *Comput. Struct. Biotechnol. J.* **2012**, *14*, 154–160.
- (44) Ullmann, K.; Poggemann, L.; Nirschl, H.; Lenewit, G. Adsorption process for phospholipids of different chain lengths at a fluorocarbon/water interface studied by Du Noüy ring and spinning drop. *Colloid Polym. Sci.* **2020**, *298*, 407–417.
- (45) Aliabouzar, M.; Lu, X.; Kripfgans, O. D.; Fowlkes, J. B.; Fabiilli, M. L. Acoustic droplet vaporization in acoustically responsive scaffolds: Effects of frequency of excitation, volume fraction and threshold determination method. *Ultrasound Med. Biol.* **2019**, *45* (12), 3246–3260.
- (46) Kaushik, A.; Abeid, B. A.; Estrada, J. B.; Fowlkes, J. B.; Fabiilli, M. L.; Aliabouzar, M. The radial dynamics and acoustic emissions of phase-shift droplets are impacted by mechanical properties of tissue-mimicking hydrogels. *Ultrason. Sonochem.* **2024**, *109*, No. 106984.
- (47) Segers, T.; Gaud, E.; Casqueiro, G.; Lassus, A.; Versluis, M.; Frinking, P. Foam-free monodisperse lipid-coated ultrasound contrast agent synthesis by flow-focusing through multi-gas-component microbubble stabilization. *Appl. Phys. Lett.* **2020**, *116* (17), 173701.
- (48) Balasco, N.; Diaferia, C.; Rosa, E.; Monti, A.; Ruvo, M.; Doti, N.; Vitagliano, L. A comprehensive analysis of the intrinsic visible fluorescence emitted by peptide/protein amyloid-like assemblies. *Int. J. Mol. Sci.* **2023**, *24* (9), 8372.
- (49) Faez, T.; Goertz, D.; De Jong, N. Characterization of Definity ultrasound contrast agent at frequency range of 5–15 MHz. *Ultrasound Med. Biol.* **2011**, *37* (2), 338–342.
- (50) Morgan, K. E.; Allen, J. S.; Dayton, P. A.; Chomas, J. E.; Klibaov, A.; Ferrara, K. W. Experimental and theoretical evaluation of microbubble behavior: Effect of transmitted phase and bubble size. *IEEE Trans. Sonics Ultrason.* **2000**, *47* (6), 1494–1509.
- (51) Hoff, L.; Sontum, P. C.; Hovem, J. M. Oscillations of polymeric microbubbles: Effect of the encapsulating shell. *J. Acoust. Soc. Am.* **2000**, *107* (4), 2272–2280.
- (52) Aliabouzar, M.; Abeid, B. A.; Kripfgans, O. D.; Fowlkes, J. B.; Estrada, J. B.; Fabiilli, M. L. Real-time spatiotemporal characterization of mechanics and sonoporation of acoustic droplet vaporization in acoustically responsive scaffolds. *Appl. Phys. Lett.* **2023**, *123* (11), 114101.
- (53) Fan, C.-H.; Lin, Y.-T.; Ho, Y.-J.; Yeh, C.-K. Spatial-temporal cellular bioeffects from acoustic droplet vaporization. *Theranostics* **2018**, *8* (20), 5731.
- (54) Qin, D.; Zhang, L.; Chang, N.; Ni, P.; Zong, Y.; Bouakaz, A.; Wan, M.; Feng, Y. In situ observation of single cell response to acoustic droplet vaporization: Membrane deformation, permeabilization, and blebbing. *Ultrason. Sonochem.* **2018**, *47*, 141–150.
- (55) Sheng, D.; Deng, L.; Li, P.; Wang, Z.; Zhang, Q. Perfluorocarbon nanodroplets with deep tumor penetration and controlled drug delivery for ultrasound/fluorescence imaging guided breast cancer therapy. *ACS Biomater. Sci. Eng.* **2021**, *7* (2), 605–616.
- (56) Mountford, P. A.; Thomas, A. N.; Borden, M. A. Thermal activation of superheated lipid-coated perfluorocarbon drops. *Langmuir* **2015**, *31*, 4627–4634.
- (57) Lacour, T.; Guédré, M.; Valier-Brasier, T.; Coulouvrat, F. A model for acoustic vaporization dynamics of a bubble/droplet system

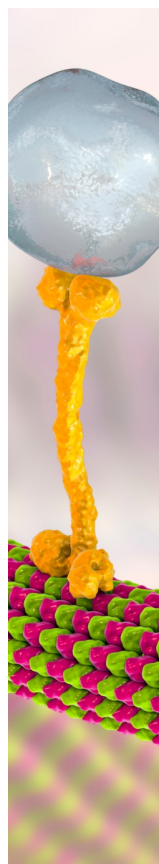
encapsulated within a hyperelastic shell. *J. Acoust. Soc. Am.* **2018**, *143* (1), 23–37.

(58) Li, D. S.; Kripfgans, O. D.; Fabiilli, M. L.; Brian Fowlkes, J.; Bull, J. L. Initial nucleation site formation due to acoustic droplet vaporization. *Appl. Phys. Lett.* **2014**, *104* (6), 387–406.

(59) Shpak, O.; Kokhuis, T. J.; Luan, Y.; Lohse, D.; de Jong, N.; Fowlkes, B.; Fabiilli, M.; Versluis, M. Ultrafast dynamics of the acoustic vaporization of phase-change microdroplets. *J. Acoust. Soc. Am.* **2013**, *134* (2), 1610–1621.

(60) Chomas, J. E.; Dayton, P.; Allen, J.; Morgan, K.; Ferrara, K. W. Mechanisms of contrast agent destruction. *IEEE Trans. Sonics Ultrason.* **2001**, *48* (1), 232–248.

(61) Aliabouzar, M.; Kripfgans, O. D.; Wang, W. Y.; Baker, B. M.; Fowlkes, J. B.; Fabiilli, M. L. Stable and transient bubble formation in acoustically-responsive scaffolds by acoustic droplet vaporization: theory and application in sequential release. *Ultrason. Sonochem.* **2021**, *72*, No. 105430.



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